



LAB PRACTICAL – 10

(Automatic Load Balancer with Auto Scaling Group With Webserver installed Instance creation and Provisioning)

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Steps to Set Up Load Balancing with EC2 Instances

Step1 : - **Launch EC2 Instances:** Create two or more EC2 instances, install a web server on each, and tag them for easy identification.

aws [Search] [Alt+S] Europe (Stockholm) hdhph

EC2 > Instances > Launch an instance

It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices. Do not show me this message again Take a walkthrough

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

My_Webserver1 Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Search our full catalog including 1000s of application and OS images

Recents My AMIs Quick Start

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

Summary

Number of instances 1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.7.2...read more
ami-0274f4b62b6ae3bd5

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of

Launch instance

EC2 > Instances > Launch an instance

Allow tags in metadata

User data - optional

Upload a file with your user data or enter it in the field.

Choose file

```
#!/bin/bash
sudo yum update -y

# Install Apache web server (httpd)

sudo yum install -y httpd
sudo systemctl start httpd
sudo systemctl enable httpd

# Create a simple HTML file to verify the web server is running, including
dynamic hostname
echo "<html><h1>Welcome to Apache Web Server on Amazon Linux -
$(hostname)</h1></html>" > /var/www/html/index.html
```

☐ User data has already been base64 encoded

Summary

Number of instances 1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.7.2...read more
ami-0274f4b62b6ae3bd5

Virtual server type (instance type)
t3.micro

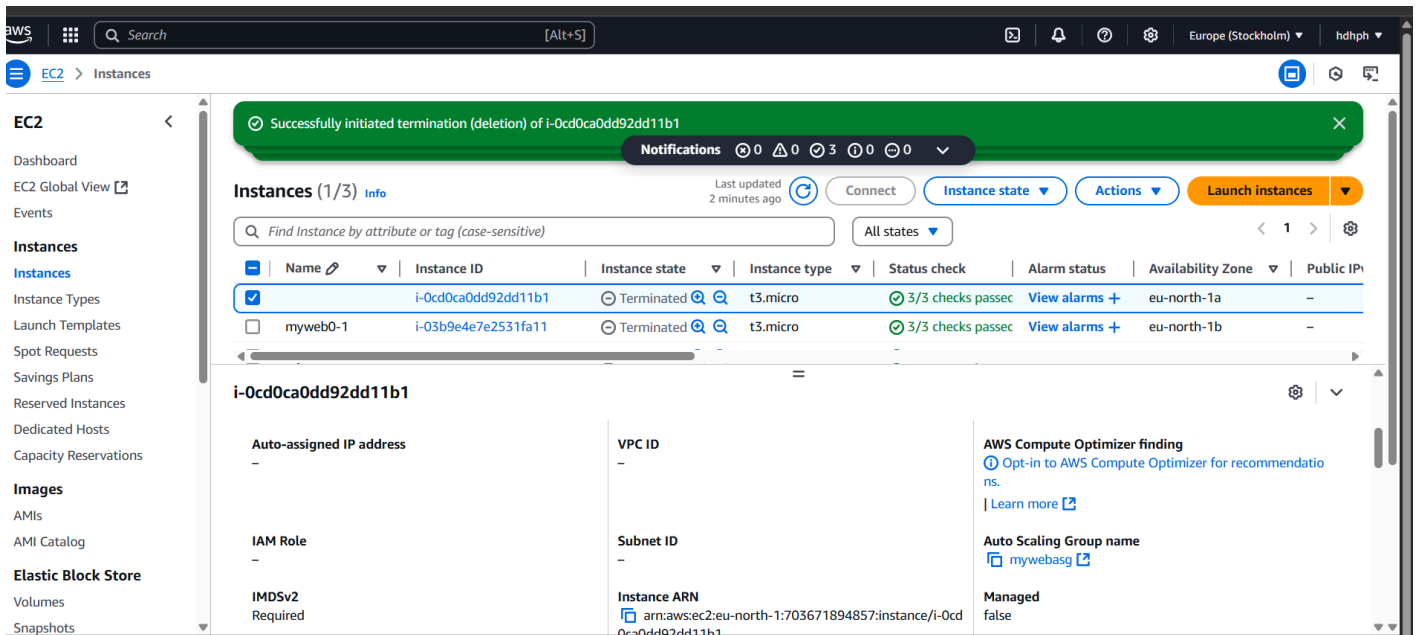
Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of

Launch instance

Preview code



Welcome to Apache Web Server on Amazon Linux - ip-172-31-41-43.eu-north-1.compute.internal!

Step 2 : Create an Application Load Balancer (ALB)

- Go to **Load Balancers > Create Load Balancer**.
- Select **Application Load Balancer**.
- Name the ALB.
- Scheme: **Internet-facing**, IP type: **IPv4**.
- Select VPC and at least 2 **availability zones/subnets**.
- Add a **listener** on port **80**.
- In Target Group section, select the one you created earlier.

- Configure security groups (allow HTTP).
- Review and create.

EC2 > Load balancers > Compare and select load balancer type

Compare and select load balancer type

A complete feature-by-feature comparison along with detailed highlights is also available. [Learn more](#)

Load balancer types

Application Load Balancer [Info](#)

Network Load Balancer [Info](#)

Gateway Load Balancer [Info](#)

aws Search [Alt+S]

EC2 > Load balancers > Create Application Load Balancer

Application Load Balancers now support public IPv4 IP Address Management (IPAM)

You can get started with this feature by configuring IP pools in the Network mapping section.

Create Application Load Balancer [Info](#)

The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances, microservices, and containers, based on request attributes. When the load balancer receives a connection request, it evaluates the listener rules in priority order to determine which rule to apply, and if applicable, it selects a target from the target group for the rule action.

► How Application Load Balancers work

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)

Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name is publicly resolvable.
- Requires a public subnet.

☐ **Internal**

- Serves internal traffic.
- Has private IP addresses.
- DNS name is publicly resolvable.
- Compatible with the IPv4 and Dualstack IP address types.

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hdhph

EC2 > Load balancers > Create Application Load Balancer

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-01b4b275a40ebe899
IPv4 subnet CIDR: 172.31.0.0/20

Security groups Info

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups

Select up to 5 security groups

default
sg-0aa81526818fe02d7 VPC: vpc-01bdc1fe8f722ba30

Listeners and routing Info

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP:80

Remove

Protocol

Port

HTTP

:

80

1-65535

Default action

Info

Forward to

Select a target group

Create target group

Step4 : - Create Security group

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EC2 > Security Groups > Create security group

Create security group Info

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name Info

mywebser-lnb-sg

Name cannot be edited after creation.

Description Info

Security Group for LB

VPC Info

vpc-01bdc1fe8f722ba30

Inbound rules Info

This security group has no inbound rules.

Add rule

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Security Groups

Capacity Reservations

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Load Balancers

Target Groups

Security Groups (3) Info

Actions

Export security groups to CSV

Create security group

Find resources by attribute or tag

	Name	Security group ID	Security group name	VPC ID	Description
☐	-	sg-0a322ac9b4aa88189	launch-wizard-1	vpc-01bdc1fe8f722ba30	launch-wizard-
☐	-	sg-0aa81526818fe02d7	default	vpc-01bdc1fe8f722ba30	default VPC sec
☐	-	sg-0352a92bd36519c8b	mywebblb	vpc-01bdc1fe8f722ba30	sg for lb

Select a security group

sg-0a322ac9b4aa88189 - launch-wizard-1

Actions

Details

Security group name

[launch-wizard-1](#)

Security group ID

[sg-0a322ac9b4aa88189](#)

Description

[launch-wizard-1](#) created 2025-04-12T09:33:25.762Z

VPC ID

[vpc-01bdc1fe8f722ba30](#)

Owner

[703671894857](#)

Inbound rules count

3 Permission entries

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing - new

VPC associations - new

Tags

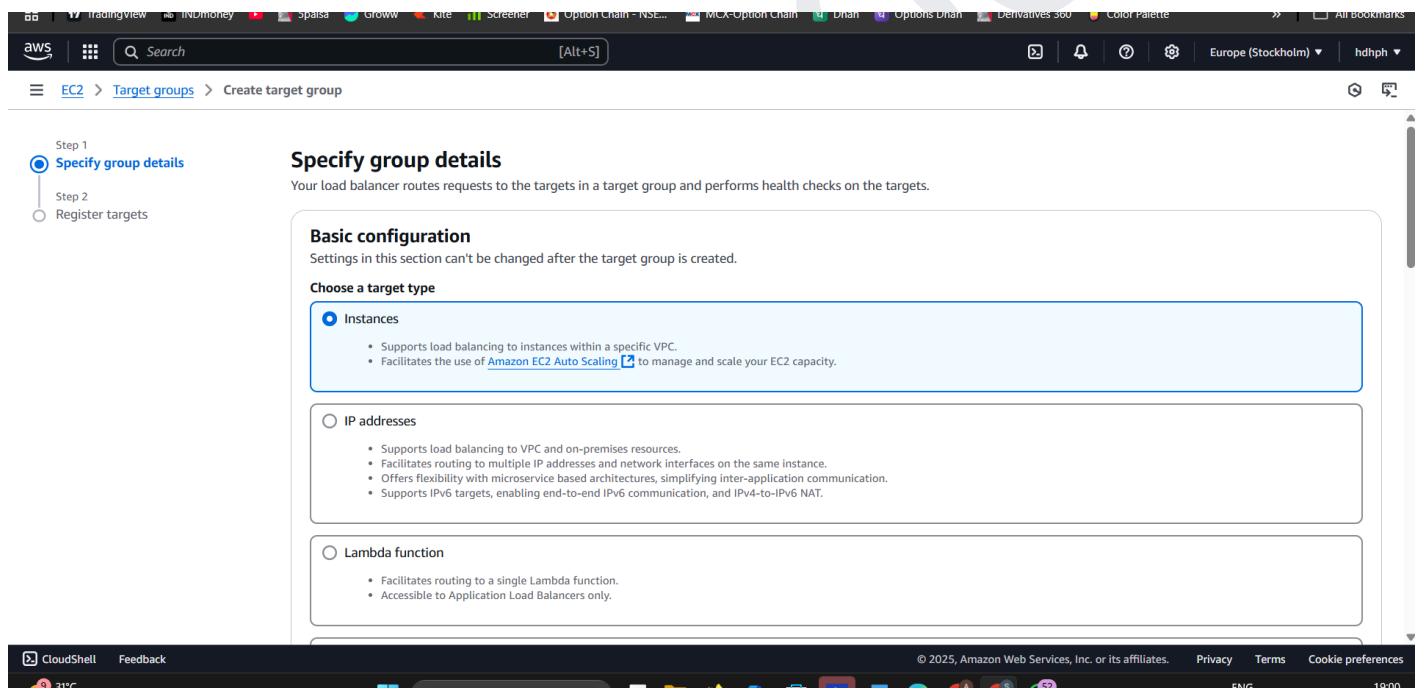
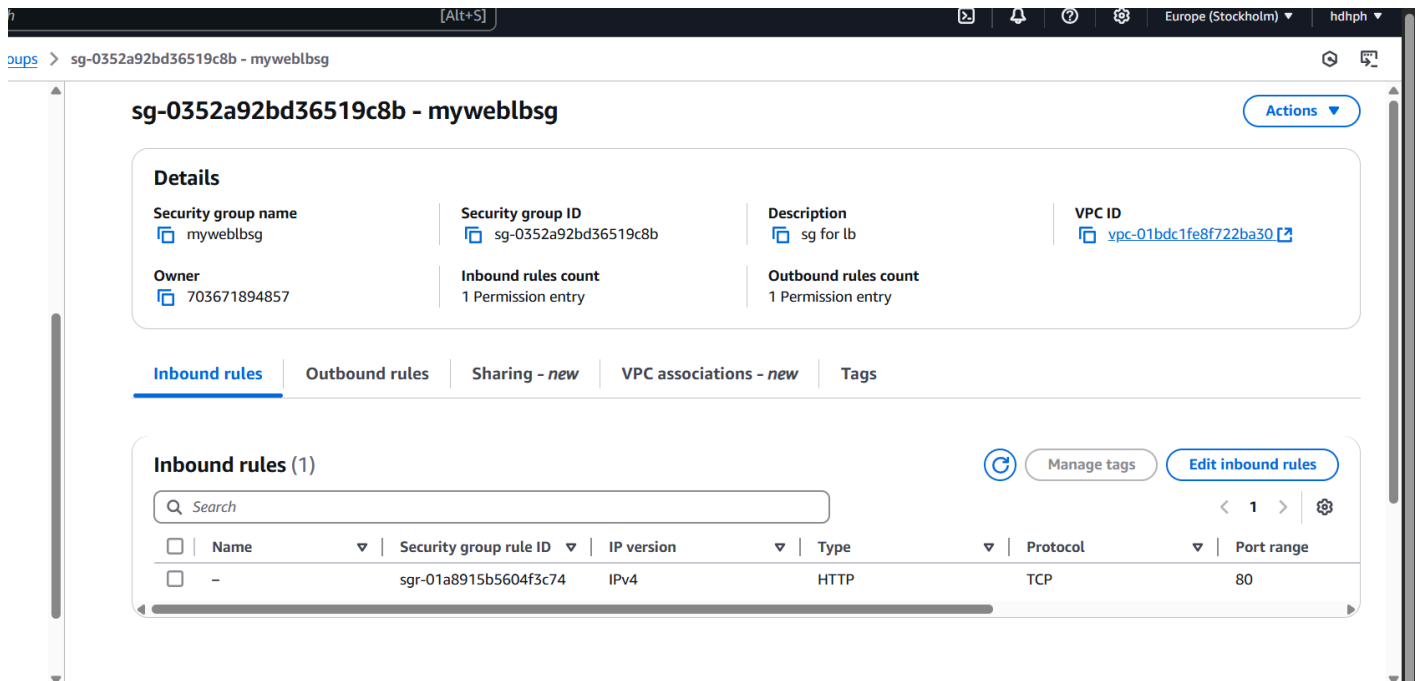
Inbound rules (3)

Manage tags

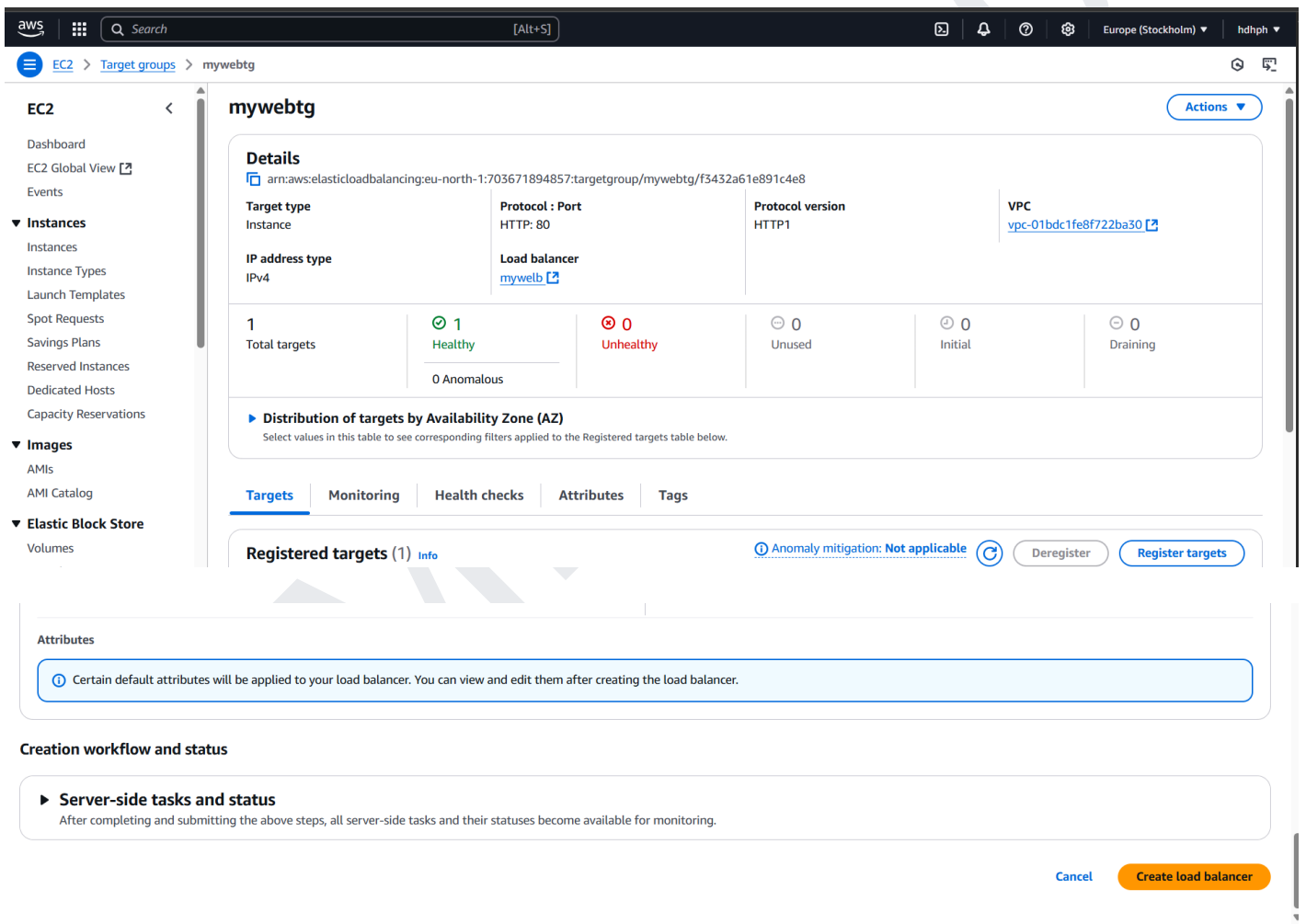
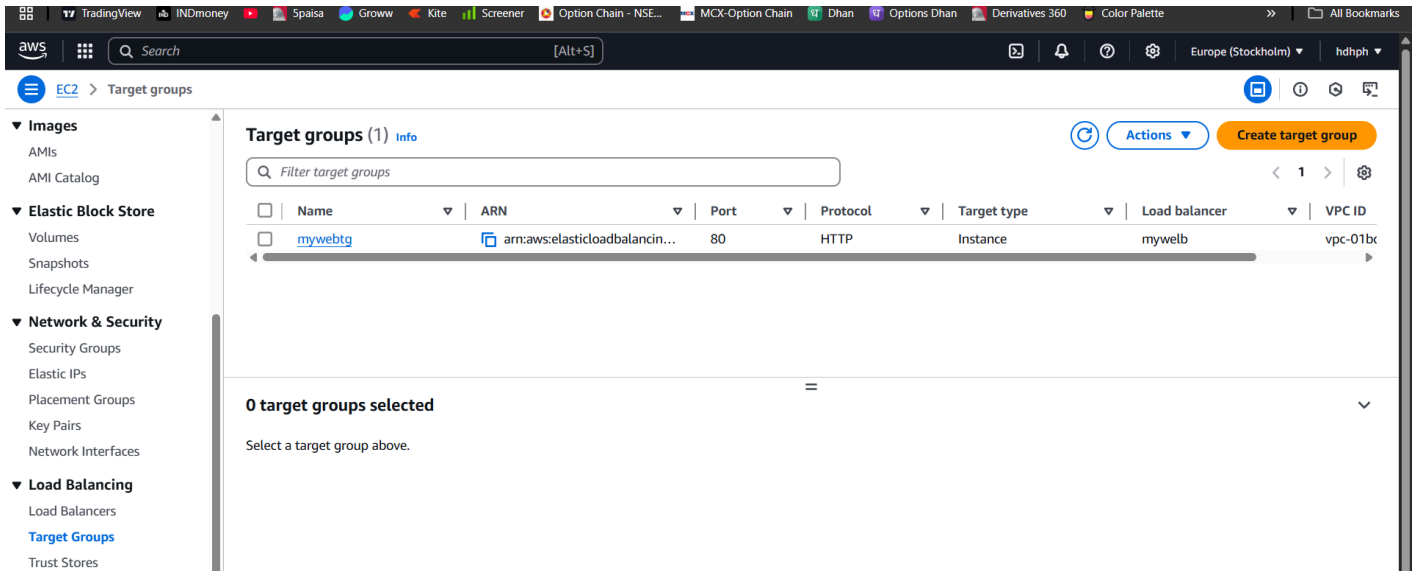
Edit inbound rules

Search

	Name	Security group rule ID	IP version	Type	Protocol	Port range
☐	-	sgr-08986658c7d06b7a6	IPv4	HTTP	TCP	80
☐	-	sgr-02aa6af14b68e4f3a	IPv4	SSH	TCP	22
☐	-	sgr-06ef7h99905483e3	IPv4	HTTPS	TCP	443



Step 5 : - Create Target group



Step 6 : - Successfully created Load balancer

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EC2 > Load balancers

Capacity Reservations

▼ Images

- AMIs
- AMI Catalog

▼ Elastic Block Store

- Volumes
- Snapshots
- Lifecycle Manager

▼ Network & Security

- Security Groups
- Elastic IPs
- Placement Groups
- Key Pairs
- Network Interfaces

▼ Load Balancing

- Load Balancers**
- Target Groups
- Trust Stores

▼ Auto Scaling

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Load balancers (1/1)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

☒	Name	DNS name	State	VPC ID	Availability Zones	Type	Date create
☒	mywelb	mywelb-1956613209.eu-no...	Active	vpc-01bdc1fe8f722ba30	3 Availability Zones	application	April 12, 20...

Load balancer: mywelb

Listeners and rules | Network mapping | Resource map | **Security** | Monitoring | Integrations | Attributes | Capacity | Tags

Security groups (1)

A security group is a set of firewall rules that control the traffic to your load balancer.

Security Group ID	Name	Description
sg-0aa81526818fe02d7	default	default VPC security group

Edit

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EC2 > Load balancers > mywelb

Application Load Balancers now support public IPv4 IP Address Management (IPAM)
You can get started with this feature by configuring IP pools in the Network mapping section.

Edit IP pools

mywelb

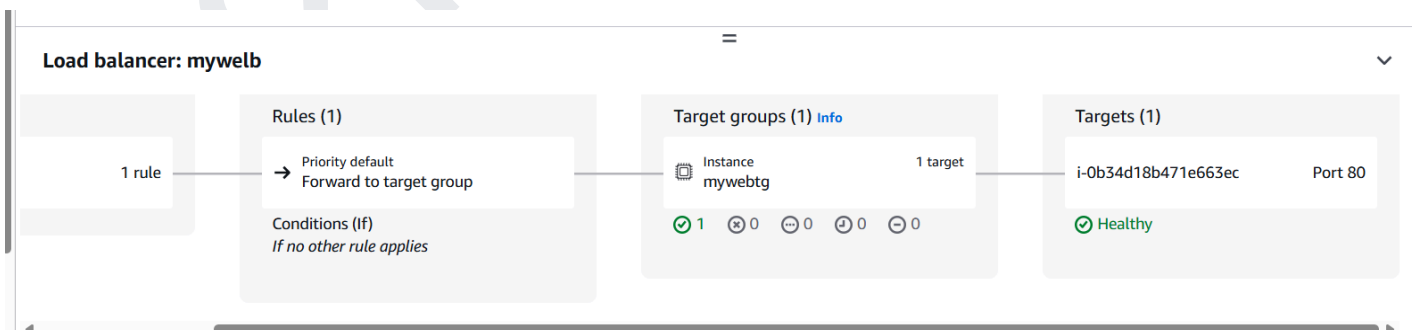
Details

Load balancer type Application	Status Active	VPC vpc-01bdc1fe8f722ba30	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone Z23TAZ6LKFMMNIO	Availability Zones subnet-0ab11901d0552efdb eu-north-1a (eun1-az1) subnet-0a44616e683781f14 eu-north-1b (eun1-az2) subnet-01b4b275a40ebe899 eu-north-1c (eun1-az3)	Date created April 12, 2025, 15:23 (UTC+05:30)
Load balancer ARN arn:aws:elasticloadbalancing:eu-north-1:703671894857:loadbalancer/app/mywelb/a15582f953ab013c		DNS name info mywelb-1956613209.eu-north-1.elb.amazonaws.com (A Record)	

Listeners and rules | Network mapping | Resource map | Security | Monitoring | Integrations | Attributes | Capacity | Tags

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Step 7 : Register Targets: Add the EC2 instances to the target group and set up health checks.

EC2 > Target groups > mywebtg > Register targets

Register targets

Select instances, specify ports, and add the instances to the list of pending targets. Repeat to add additional combinations of instances and ports to the list of pending targets. Once you are satisfied with your selections, click Register pending targets.

Available instances (1)

Filter instances

< 1 > ⚙

☐	Instance ID	Name	State	Security groups	Zone	Private IPv4 address
☐	i-0b34d18b471e663ec		Running	mywebtblsg	eu-north-1a	172.31.17.185

0 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

Include as pending below

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1-65535 (separate multiple ports with commas)

Include as pending below

Review targets

Targets (1)

Filter targets

Show only pending

Remove all pending

< 1 > ⚙

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
i-0b34d18b471e663ec		80	Running	mywebtblsg	eu-north-1a	172.31.17.185	subnet-0ab11901d0552efdb	April 12, 2025, 18:41 (UTC+05:30)

0 pending

Cancel

Register pending targets

Welcome to Apache Web Server on Amazon Linux - ip-172-31-41-43.eu-north-1.compute.internal!

Welcome to Apache Web Server on Amazon Linux - ip-172-31-41-43.eu-north-1.compute.internal!

Step 7 :- Create Auto Scaling Group (ASG)

- Go to **Auto Scaling Groups > Create ASG**.
- Choose **Launch Template** created in step 1.
- Name your ASG.
- Select **VPC and Subnets** (multiple AZs recommended).
- Attach **Application Load Balancer** and previously created **Target Group**.
- Configure **Health Check Type** as **EC2 + ELB**.
- Set **desired, min, and max** number of instances.
- Add **Scaling Policies**:
 - Target tracking (e.g., keep CPU utilization at 50%) or Step Scaling.
- (Optional) Add notifications to receive alerts (email, SNS).
- Review and Create the ASG.

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EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1

Choose launch template

Step 2

Choose instance launch options

Step 3 - optional

Integrate with other services

Step 4 - optional

Configure group size and scaling

Step 5 - optional

Add notifications

Step 6 - optional

Add tags

Step 7

Review

Choose launch template [Info](#)

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.

Name

Auto Scaling group name
Enter a name to identify the group.

Must be unique to this account in the current Region and no more than 255 characters.

Launch template [Info](#)

For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template
Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

[Create a launch template](#)

Cancel

Next

EC2 > Launch templates > Create launch template

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - *required*

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '!', '!', '@'.

Template version description

Max 255 chars

Auto Scaling guidance [Info](#)
Select this if you intend to use this template with EC2 Auto Scaling

☒ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

Template tags

Source template

Summary

Software Image (AMI)

-

Virtual server type (instance type)

-

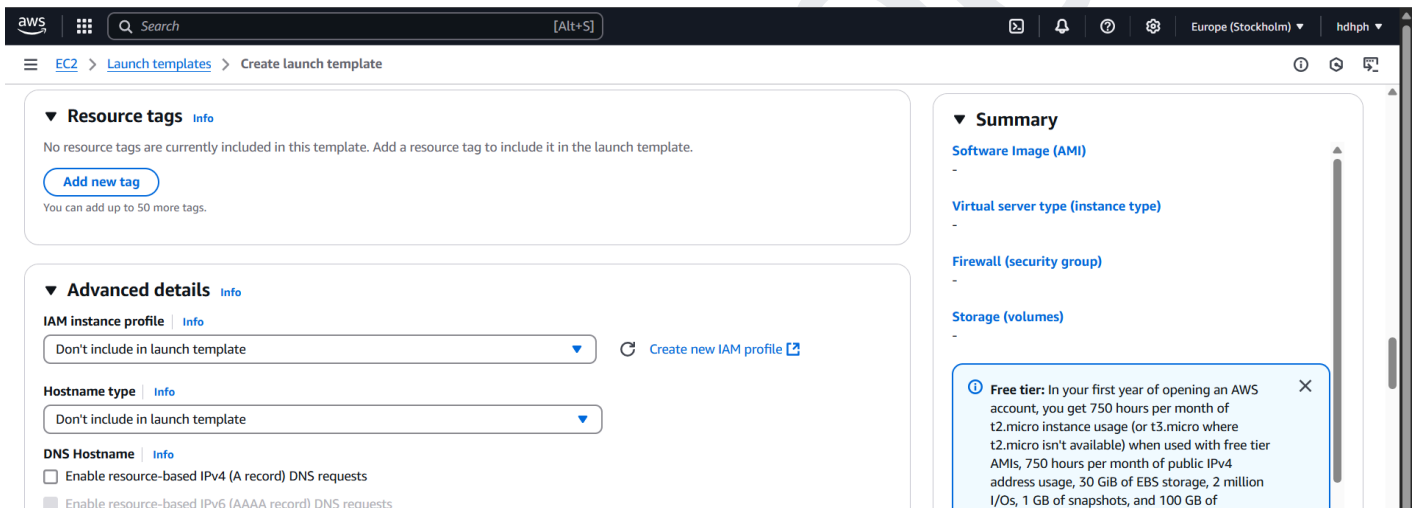
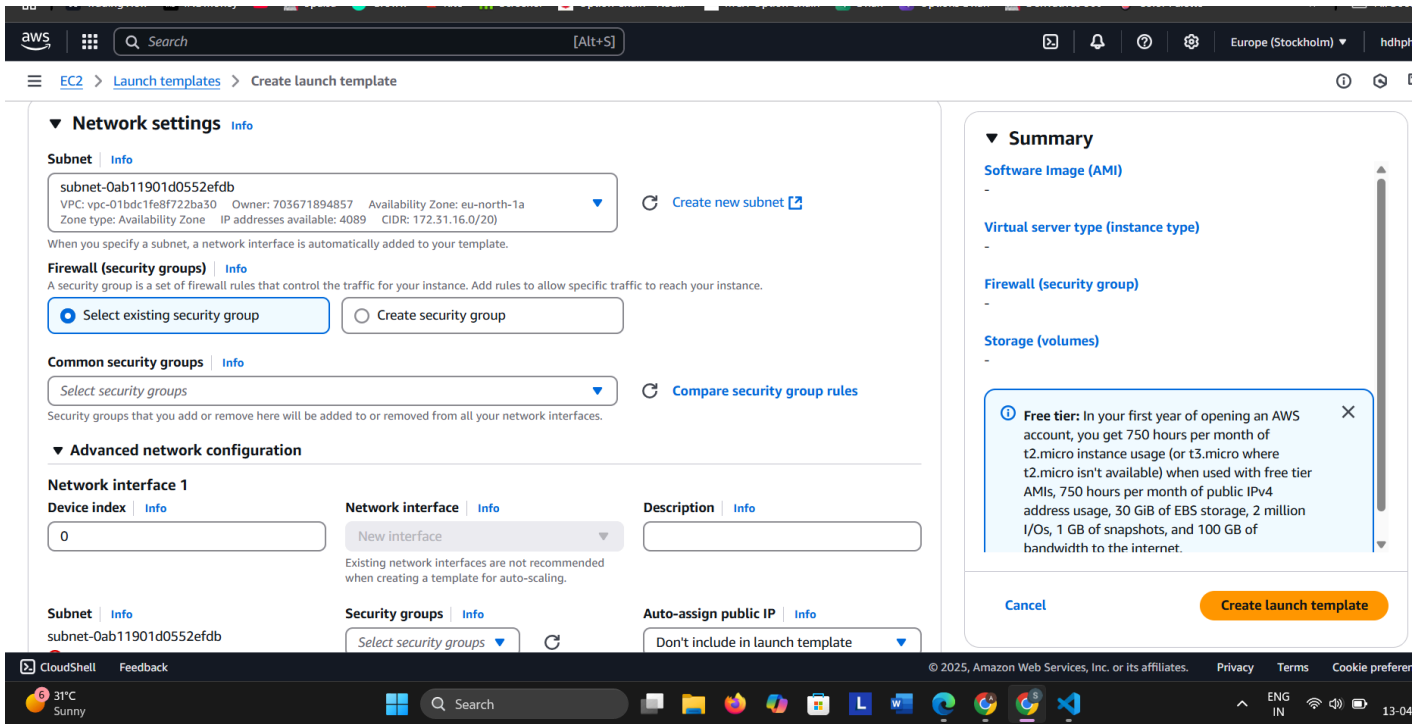
Firewall (security group)

-

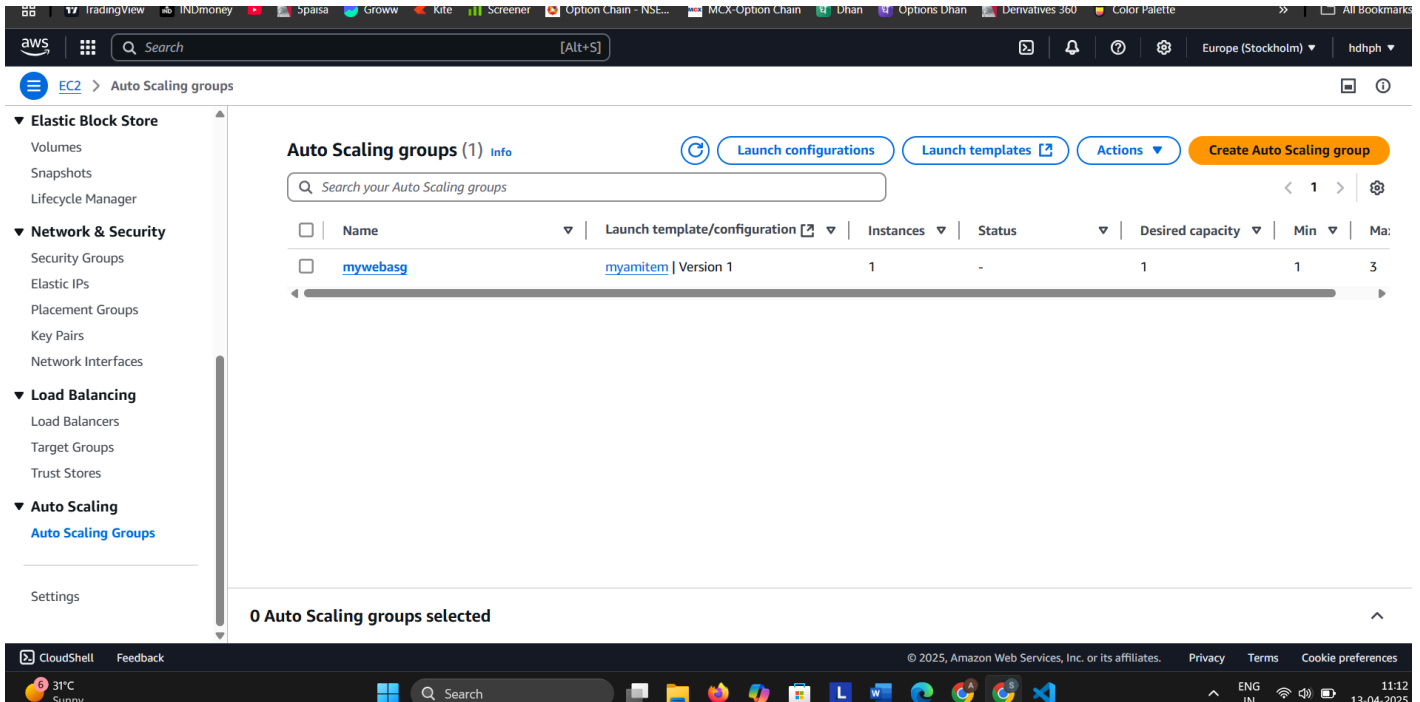
Storage (volumes)

-

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

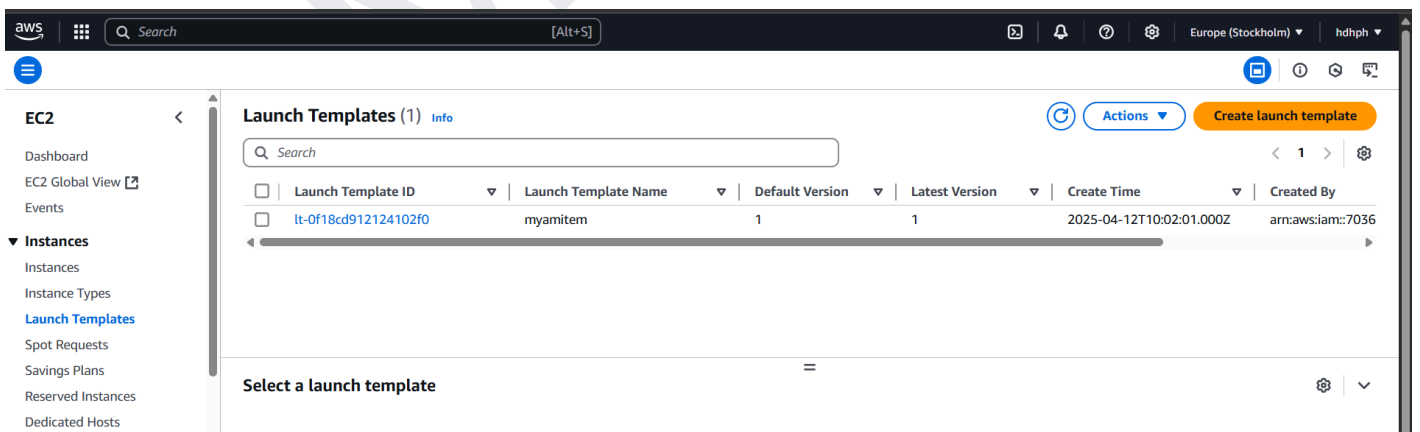


Step 8 : Successfully Created Auto scaling group



Step 9. Verify the Setup

- Visit the **DNS of the Load Balancer** (found in the Load Balancer details).
- You should see the message:
“Deployed via Auto Scaling Group”
from the web server.
- Test scaling by manually increasing load or adjusting desired capacity.



plates > myamitem

myamitem (lt-0f18cd912124102f0)

ActionsDelete template

Launch template details

Launch template ID
lt-0f18cd912124102f0

Launch template name
myamitem

Default version
1

Owner
arn:aws:iam::703671894857:root

Details

Versions

Template tags

Launch template version details

Version
1 (Default)

Description
-

Date created
2025-04-12T10:02:01.000Z

Created by
arn:aws:iam::703671894857:root

Instance details

Storage

Resource tags

Network interfaces

Advanced details

AMI ID
ami-05b4ba47bbaf8f463

Instance type
t3.micro

Availability Zone
-

Key pair name
-

Security groups
-

Security group IDs
sg-0352a92bd36519c8b

Welcome to Apache Web Server on Amazon Linux - ip-172-31-41-43.eu-north-1.compute.internal!

